

Tutorial 7

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Time dependent perturbation theory.

$$H = H^0 + \epsilon H'(t) \quad (1)$$

2 level system

$$\psi(t) = c_a(t) \psi_a e^{-\frac{i\epsilon_a t}{\hbar}} + c_b(t) \psi_b e^{-\frac{i\epsilon_b t}{\hbar}} \quad (2)$$

Time dependent Schrodinger equation.

$$H\psi = i\hbar \frac{\partial \psi}{\partial t} \quad (3)$$

with $H_{a,a'} = H_{b,b'} = 0$. $\omega_0 = \frac{\epsilon_b - \epsilon_a}{\hbar}$.
1st order (assume $c_a(0) = 1$, $c_b(0) = 0$).

$$c_a^j(t) = 1 \quad (4)$$

$$\frac{dc_b^1}{dt} = -\frac{i}{\hbar} H'_{ba} e^{i\omega_0 t} \quad (5)$$

Question 1

$$H = H_0 + \epsilon V(x, t) \quad (6)$$

given

$$V(x, t) = \begin{cases} x - \frac{a}{2} & \text{if } 0 < t < \tau \\ 0 & t > \tau \end{cases} \quad (7)$$

Probability to be in 1st excited state for large t ?

(Can consider a two level system).

Unperturbed

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad (8)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad (9)$$

$$\psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \quad (10)$$

$$\psi_2(x) \sin\left(\frac{2\pi x}{a}\right) \quad (11)$$

$$H_{a,a'} = \langle \psi_1 | H | \psi_1 \rangle \propto \int_0^a \left(\sin\left(\frac{\pi x}{a}\right) \right)^2 \left(x - \frac{a}{2} \right) dx = 0 \quad (12)$$

$$H'_{ba} = \langle \psi_b | H(t) | \psi_a \rangle = \int_0^a \psi_2(x) \left(x - \frac{a}{2} \right) \psi_1(x) dx = \frac{2}{a} \int_0^a \sin\left(\frac{2\pi x}{a}\right) dx \quad (13)$$

$$= -\frac{16a}{9\pi^2} \quad (14)$$

(no time dependence) we can use (5)

$$c_2(\tau) = \frac{16a}{9\pi^2\hbar} \int_0^\tau dt e^{i\omega_{21}t} = \frac{16a}{9\pi^2\hbar} \frac{1}{\omega_{21}} \left(e^{i\omega_{21}\tau} - 1 \right) \quad (15)$$

constant in time if $t > \tau$.

Transition frequency

$$\omega_{21} = \frac{E_2 - E_1}{\hbar} = \frac{3\pi^2\hbar}{2ma^2} \quad (16)$$

Probability for the first excited state. $|c_2(\tau)|^2 = P_{1 \rightarrow 2} \propto a \sin^2(b\omega_{21}\tau)$. Purely QM phenomena. Oscillatory behavior P can vary a lot.